Exp No: 25

PRODUCER–CONSUMER PROBLEM USING SYSTEMC

AIM:

To implement and simulate the Producer–Consumer problem using SystemC and achieve synchronization between concurrent processes using an FIFO channel.

PROCEDURE:

1. Create Producer and Consumer modules using SC_MODULE in SystemC.
2. Create a FIFO channel of fixed size and connect the producer output and consumer input to it.
3. Implement the producer process using SC_THREAD to generate and write five integer values into the FIFO.
4. Implement the consumer process using SC_THREAD to read values from the FIFO and display them.
5. Compile and execute the program, run the simulation for 10 seconds, and observe the synchronization between the producer and consumer.

PROGRAM:

```
#include <systemc.h>


// Producer Module

SC_MODULE(Producer) {

sc_fifo_out<int> out;


void produce() {

for (int i = 1; i <= 5; i++) {

cout << "Produced: " << i << endl;

out.write(i);

wait(1, SC_SEC);


}

}
```

```
SC_CTOR(Producer) {
SC_THREAD(produce);
}
};

// Consumer Module
SC_MODULE(Consumer) {
sc_fifo_in<int> in;

void consume() {
int value;
while (true) {
in.read(value);
cout << "Consumed: " << value << endl;
wait(1, SC_SEC);
}
}

SC_CTOR(Consumer) {
SC_THREAD(consume);
}
};

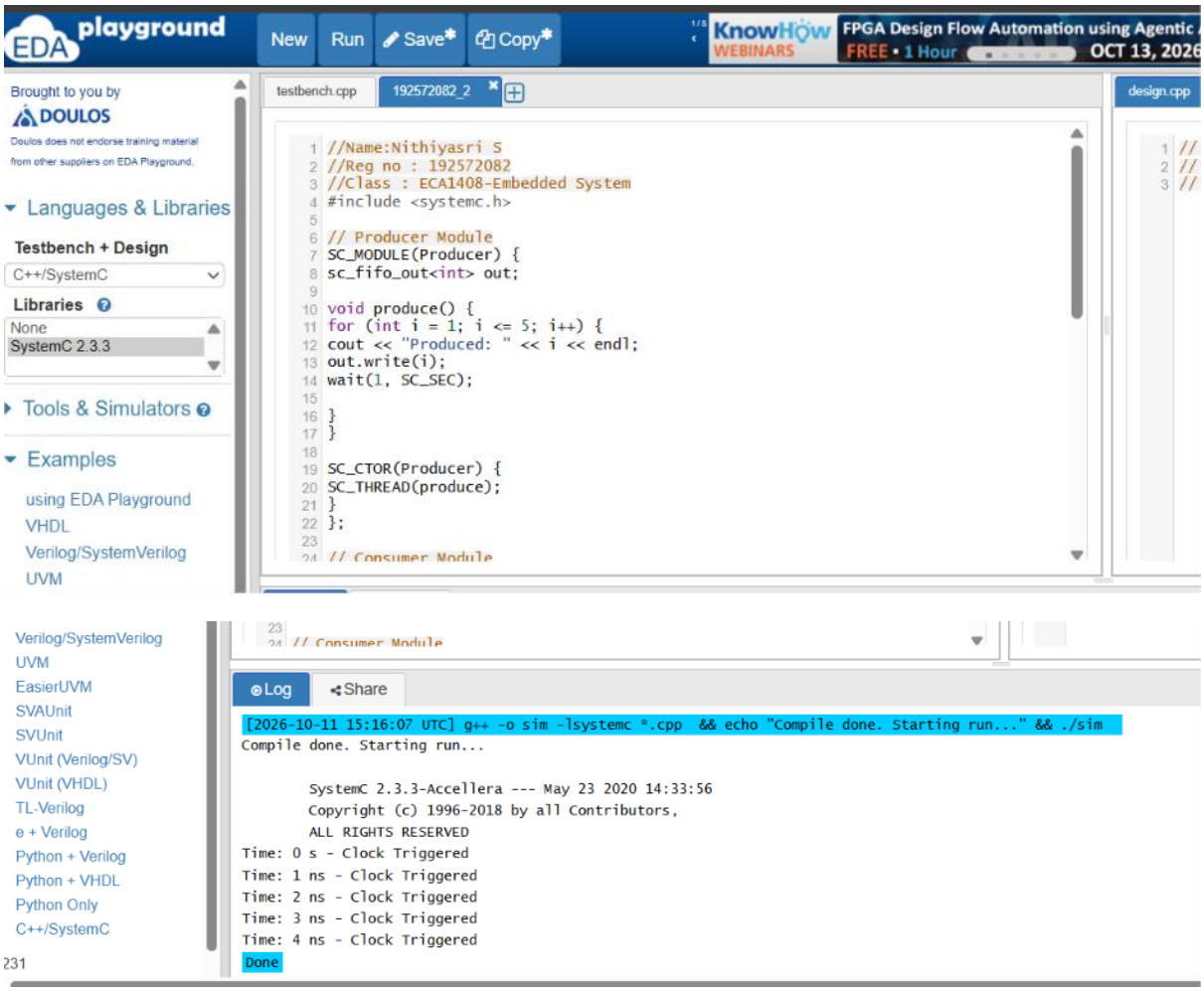
int sc_main(int argc, char* argv[]) {
sc_fifo<int> fifo(5);

Producer producer("Producer");
Consumer consumer("Consumer");
```

```
producer.out(fifo);
consumer.in(fifo);

sc_start(10, SC_SEC);

return 0;
}
```



RESULT:

Thus, the Producer–Consumer problem was successfully implemented and simulated using SystemC, achieving synchronization between concurrent processes through FIFO communication.